

Atoms, Aggregates, and Elementary Predicates: on Proper Names in Italian and beyond

Abstract

This article challenges uniform predicativism and covert light noun analyses of proper names. Using Italian and cross-linguistic data, we argue that onomastic morphosyntax relies on a mereosyntactic dichotomy: roots conceptualized as atomic individuals merge in D as referential expressions, whereas aggregates merge in N, requiring functional support. This distinction explains asymmetries in toponyms, temporal names, and plural names. For complex names (e.g., *Lago di Garda*), we propose they avoid ordinary predication by recruiting the exact mereological relators used for material constitution, directly fusing a sortal noun with a rigid designator into a unified domain.

Keywords: Proper Names, Complex Names, Light Noun Hypothesis, Italian, Mereology

1. Introduction

At the interfaces of syntax, semantics, and the philosophy of language, the categorical status of proper names remains a subject of intense and unresolved debate. Historically, the *orthodox* view — rooted in Kripkean semantics (Kripke 1980) — has treated proper names as rigid designators and inherently singular referential terms. However, this strictly referentialist approach has been challenged by Predicativism (e.g., Burge 1973; Matushansky 2008; Fara 2015, among others). Predicativists argue that proper names are uniformly predicates (i.e., common count nouns) whose semantic value is the property of "being called N".¹ To account for the use of proper names as singular terms in argument position, standard versions of Predicativism postulate that bare names are actually embedded within a concealed determiner phrase headed by an unpronounced definite article ($\emptyset$ *the*).

While Predicativism offers a uniform semantic analysis, it has encountered substantial empirical and theoretical criticism. A growing number of scholars (Leckie 2013; Rami 2014, 2015; Jeshion 2015, 2017; Schoubye 2017; Delgado 2019, 2022) have extensively documented that proper names and common count nouns exhibit radically different morphosyntactic properties. As

Delgado (2022) and Jeshion (2017) point out, the postulation of an unpronounced determiner does not unify proper names and count nouns; Rather, it renders proper names exceptional as count nouns, imposing an anomalous syntactic distribution upon them. Recently, Moltmann (2022) argues for the non-uniformity of proper names. Challenging the assumption that proper names are uniformly count nouns, she argues, with her Light Noun Hypothesis, that simple names divide into mass and count depending on a covert functional structure. Specifically, she claims that "Type 1" names (anthroponyms and toponyms) combine with silent light nouns functioning as classifiers (e.g., PERSON, PLACE, THING), which entirely determine their countability status. By contrast, "Type 2" names (such as rivers, mountains, and regions) combine with an overt or silent specific sortal noun and obligatorily select a definite determiner.

In this paper, we challenge the Light Noun Hypothesis and propose a different account of proper names. Drawing primarily on Italian data, we demonstrate that postulating silent light nouns like PLACE or PERSON is both empirically inadequate and theoretically unnecessary to derive the countability and the determinational needs of proper names. Instead, we argue that the morphosyntax of proper names is strictly governed by a structural dichotomy rooted in the mereological conceptualization of the referent: individuals (atoms) versus sets (aggregates).

Building on Longobardi (1994, 2005), and Franco & Rugna (2024), we show that roots conceptualized as punctual individuals (e.g., anthroponyms, cities)—lacking proper internal parts within the relevant semantic domain—bypass the N-layer entirely and undergo Merge directly in D (or raise to D), accounting for their rejection of determiners.² Conversely, roots conceptualized as aggregates or collections of weakly differentiated parts (e.g., nations, plural names, and names of musical bands) possess internal mereological complexity. They must therefore undergo Merge in N, lacking intrinsic referential autonomy, and obligatorily requiring a functional D-layer.

Furthermore, we address the issue of "complex names" (e.g., *Lago di Garda*, *Mar Mediterraneo*) by adopting the framework of Manzini and colleagues (Manzini & Savoia 2011; Manzini & Franco 2016; Franco & Manzini 2017; Rugna & Franco 2022, among others), which treats adpositions as

elementary mereological predicates. We propose that Italian employs, for complex names, the exact same relational primitives used to encode material constitution (cf. Jackendoff 1991). The paper is organized as follows. Section 2 challenges the Light Noun Hypothesis by analyzing the split in Italian toponyms and the absence of uniform spatial classifiers. Section 3 introduces evidence from plural proper names and group names to formalize the structural distinction between Merge in D (atoms) and Merge in N (aggregates). Section 4 outlines the mereological morphosyntax of complex names. Section 5 evaluates the implications of our model for the broader Predicativism vs. Referentialism debate and propose a syntactic typology of proper names in Italian. Section 6 provides typological evidence confirming that the non-uniformity of proper names is a robust cross-linguistic fact, along the line of the individual/aggregate divide. Section 7 concludes.

2. Challenging the Light Noun Hypothesis: Spatial and Temporal Asymmetries

In her recent and influential analysis of German, Moltmann (2022) challenges the traditional view of proper names as a morphosyntactically uniform category. She proposes a fundamental bifurcation between what she terms "Type 1" and "Type 2" names. According to her framework, Type 1 names — which encompass anthroponyms, toponyms, and "productive" names denoting times and numbers — are structurally characterized by the presence of a covert light noun (Kayne 2005, 2010) functioning as a classifier (e.g., PERSON, PLACE, THING). Conversely, Type 2 names — such as those designating mountains, lakes, and temples — lack this light noun projection. Instead, they obligatorily merge with a (covert or overt) specific sortal noun (e.g., *Berg* 'mountain') and a matching definite determiner.

A crucial aspect of Moltmann's proposal is that the internal morphosyntactic behavior of Type 1 names is dictated by the specific semantic properties of the selected light noun, particularly regarding its countability. Under her account, while the covert classifier PERSON is inherently a count noun, classifiers like PLACE or THING exhibit mass-like, non-countable properties. In

German, this alleged mass/count dichotomy manifests in two primary morphosyntactic diagnostics: the selection of relative pronouns and the availability of plural anaphora.

First, human-denoting proper names (Type 1 with PERSON) select *d-pronouns* (e.g., *der*, *die*, *das*), as shown in (1), whereas location-denoting proper names (Type 1 with PLACE) obligatorily select *w-pronouns* (e.g., *was*), as illustrated in (2) (adapted from Moltmann 2022: 131-134).

- (1) a. Maria, *die* /**was*
 Maria who.D /what.W
 ‘Maria, who’
- b. Fritz, *der* /**was*
 Fritz who.D /*what.W
 ‘Fritz, who’
- (2) a. München, *was* /???*das* ich sehr gut kenne
 Munich what.W / that.D I very well know
 ‘Munich, which I know very well’
- b. Ich liebe Italien, *was* /???*das* dir ja auch gut gefällt.
 I love Italy what.W / that.D you PART too well pleases
 ‘I love Italy, which pleases you too.’

Second, when conjoined, names of individuals license plural anaphoric pronouns (e.g., *sie* ‘they’), as in (3). In stark contrast, conjoined German toponyms generally fail to support plural anaphora, requiring instead the repetition of a full sortal noun phrase (e.g., *diese Städte* ‘those cities’), as in (4).

- (3) Hans und Maria... *sie* /**diese* Personen
 Hans and Maria... they /*those persons

- (4) München und Berlin... diese Städte /???sie
Munich and Berlin... those cities / ???they

According to Moltmann (2022), these structural asymmetries stem from whether the nominals function semantically as mass or count nouns. Specifically, a nominal's lack of countability is what drives its preference for certain *w*-pronouns and its inability to take plural anaphora. This uncountable characteristic is directly tied to the underlying semantics of the specific "light noun" that merges with the proper name. As she notes, the light noun "PERSON [...] classifies as count, but not so THING, TIME, and PLACE" (Moltmann 2022: 128). Therefore, the specific morphosyntactic patterns exhibited by toponyms arise because they incorporate the unpronounced, noncountable light noun PLACE. Anthroponyms display a diverging pattern because they are structured with the countable light noun PERSON. In (5) we illustrate the resulting syntactic and semantic contrasts between these categories of Type 1 names:

- (5) a. [[Hans [_{PERSON}]_{Nlight}]_{NlightP}]_{DP} *countable name*
b. [[Berlin [_{PLACE}]_{Nlight}]_{NlightP}]_{DP} *'mass' name*

On the other hand, Type 2 names — such as those designating mountains, volcanos, lakes, and temples — exhibit a completely different structural architecture. According to Moltmann, these names entirely lack a light noun projection. Instead, they obligatorily merge with a specific, full sortal noun (e.g., *Berg* 'mountain', *See* 'lake'), which can be either overt or phonologically silent. Because these sortal heads are standard count nouns, Type 2 names inherently classify as count: they consistently select *d*-pronouns, support plural anaphora, and obligatorily require an overt definite determiner that matches the sortal's features. The examples in (6) illustrate German names for mountains featuring an overt sortal, where the definite article overtly matches the sortal's gender.

- (6) a. der Mont Blanc, der Berg
 DEF.M.SG Mont Blanc DEF.M.SG mountain
 ‘Mont Blanc, the mountain’
- b. die Zugspitze, die Spitze
 DEF.F.SG Zugspitze DEF.F.SG peak
 ‘Zugspitze, the peak’
- c. das Erzgebirge, das Gebirge
 DEF.N.SG Erzgebirge DEF.N.SG mountain.range
 ‘The Ore Mountains, the mountain range’

Crucially, Moltmann extends this analysis to Type 2 names where no sortal is phonologically expressed. In such cases, the gender of the obligatory definite determiner serves as a morphological footprint, aligning with the gender of the silent sortal. For instance, the masculine article preceding the names of volcanos in (7) would be triggered by the silent presence of the masculine German sortal *Berg* ('mountain'):

- (7) a. der Fujiyama
 DEF.M.SG Fujiyama
 ‘Mount Fuji’
- b. der Vesuv
 DEF.M.SG Vesuvius
 ‘Mount Vesuvius’
- c. der Etna
 DEF.M.SG Etna
 ‘Mount Etna’

To formally capture this, Moltmann posits that Type 2 names are underlyingly compounds, in which the full sortal noun — whether overt or covert — acts as the syntactic head, as illustrated in (8):

- (8) a. der [Fujiyama [Berg]_N]_{NP}
b. der [Fujiyama [e]_N]_{NP}

However, Moltmann herself acknowledges a theoretical issue here. The culture-specificity and high lexical content of the sortals involved in Type 2 names make it difficult to treat them as semantically "light nouns". This poses a challenge to Kayne's (2005, 2010) well-established generalization that only light nouns — not full lexical nouns — can remain silent without a linguistic antecedent.

Still, there are more cross-linguistic challenges for this kind of syntax of proper names. As we intend to show, postulating silent light nouns is empirically problematic and theoretically superfluous. We will demonstrate that: (i) robust morphosyntactic splits occur within the very same semantic class of names (e.g., within toponyms); (ii) countability properties can be derived purely structurally (i.e., *Merge in D* vs. *Merge in N*), rendering the requirement for a light noun unnecessary; and (iii) proper names denoting sets or aggregates are treated as collective noun phrases, triggering plural agreement on the finite verb and strictly requiring a definite determiner.

In particular, a closer morphosyntactic examination of spatial and temporal proper names in Italian challenges the empirical adequacy and theoretical necessity of the Light Noun Hypothesis. Italian data reveal sharp structural splits within the same semantic class that are entirely independent of lexical roots or light nouns. Instead, countability and referentiality are structurally derived from the syntactic configuration — specifically, the properties of the D-layer and, as we

will show in section 4, the elementary adpositional relators involved in the formation of ‘complex names.’

If the morphosyntactic behavior of place names were universally governed by a covert PLACE classifier, one would expect a uniform syntactic distribution within this class. Italian data, however, reveal a sharp morphosyntactic split within the very category that should uniformly select PLACE. As documented in Rizzi (1988) and further developed by Franco & Lorusso (2019), Italian locative adpositions are sensitive to the topological and cognitive nature of the location. Toponyms denoting cities, villages, and small islands obligatorily select the preposition *a* ('at/to') and strictly reject the definite determiner. Conversely, toponyms denoting countries, continents, and large islands select the preposition *in* ('in') and obligatorily require a definite determiner (cf. Franco & Lorusso 2019: 284).

- (9) a. Gianni va *a*/?*in* Milano /Pantelleria.
 Gianni go.3SG to Milan /Pantelleria
 ‘Gianni goes to Milan / to Pantelleria.’
- b. Gianni va **a/in* Italia / Sicilia.
 Gianni go.3SG to Italy / Sicily
 ‘Gianni goes to Italy / to Sicily.’
- c. ØMilano ØPantelleria vs. l’Italia la Sicilia

This asymmetry has theoretical consequences. The locations in (9a) function akin to individual personal names: they do not tolerate a determiner. Following Longobardi (1994, 2005), this indicates that they would undergo N-to-D movement (or are directly merged in D), acquiring an object-referring, punctual status. In contrast, the locations in (9b) behave as aggregate-denoting common nouns that require a D-layer to serve as an argument. A country or a large island is

conceptually and syntactically treated as an extensive aggregate of sub-locations, allowing for partitioning (e.g., *Il Portogallo settentrionale*, 'Northern Portugal').

This state of affairs is incompatible with the Light Noun Hypothesis. If Moltmann's analysis were correct, we would be forced to stipulate that the silent light noun PLACE is selected exclusively by the individual-like toponyms in (9a) to account for their lack of D. Conversely, the aggregate-like toponyms in (9b), which require an overt D, should align with her "Type 2" names and combine with an overt sortal noun. However, this is impossible in Italian: compounding a sortal noun with a country or island name is ungrammatical (**Lo stato Portogallo*, 'The state Portugal'; **L'isola Corsica*, 'The island Corsica'). The presence or absence of D in Italian toponyms is therefore not derived from covert light nouns, but from the structural position of the root (Merge in D for atomic individuals vs. Merge in N for aggregates). We assume that if the Light Noun Hypothesis were a genuine aspect of Universal Grammar, the covert classifiers PERSON and PLACE should yield cross-linguistically uniform predictions.

The inadequacy of the Light Noun Hypothesis becomes even more evident when we shift from the spatial to the temporal domain. According to Moltmann (2022), temporal names (e.g., years, days of the week, etc.) belong to Type 1 names and contain the silent light noun TIME, which is semantically and syntactically mass/non-count. However, an analysis of names of days of the week in Italian (Franco & Lorusso 2022a) has shown that these temporal items display a complex aspectual split that correlates directly with the presence of functional heads (D or P), rather than with an inherent mass property of a light noun. When names of days appear as adverbials modifying a verbal event, their interpretation radically shifts between a punctual (bounded) reading and a habitual (unbounded/iterative) reading, depending entirely on their morphosyntactic encoding. Consider the examples in (10).

- (10) a. Vado al cinema lunedì.
 go.1SG to.DEF.M.SG cinema Monday

‘I will go to the cinema next Monday.’ (Punctual reading)

- b. Vado al cinema il lunedì.
 go.1SG to.DEF.M.SG cinema DEF.M.SG Monday

‘I (usually) go to the cinema on Mondays.’ (Habitual reading)

- c. Vado al cinema di lunedì.
 go.1SG to.DEF.M.SG cinema of Monday

‘I (usually) go to the cinema on Mondays.’ (Habitual reading)

In (10a), the proper temporal name is 'bare'. Exactly like the individual toponyms in (9a), bare temporal names undergo N-to-D movement. The N-to-D chain triggers an individual-like, bounded reference, forcing the event to be perceived as strictly punctual (a specific, deictic temporal target). If the temporal root *lunedì* inherently merged with a noncount/mass light noun (TIME), as Moltmann suggests, it should not be able to systematically act as a strict punctual, countable individual. Instead, the habitual (plural/mass-like) reading is actively derived by preventing N-to-D movement. In (10b), the presence of the definite determiner *il* keeps the nominal root in N. Here, the article functions as a generic operator (cf. Zamparelli 2000) introducing a class of temporal intervals (the set of Mondays), yielding a habitual interpretation. Even more striking is the pattern in (10c), where the proper temporal name is introduced by the preposition *di* ('of'). Building on Manzini & Savoia (2011), we assume that *di* spells out an elementary inclusion/part-whole relator. In this configuration, the bare temporal NP (*lunedì*), lacking a D superstructure, denotes a mass-like set (the class of Mondays). The relator *di* structurally maps the eventive predicate as being included within this temporal class, obligatorily forcing an anti-individuation effect and, consequently, a habitual/iterative reading.

The Italian temporal data thus weaken the Light Noun Hypothesis. A root like *lunedì* is not inherently specified as mass or count by a silent light noun. Rather, it is the syntactic derivation that determines its referential and aspectual properties: N-to-D movement yields a punctual, individual

reference (Merge in D), whereas embedding the nominal under a D operator or a prepositional part-whole relator treats the root as a set of intervals, yielding a habitual reading (Merge in N).

Our individual/aggregate name dichotomy also accounts for the German data that Moltmann (2022) uses to motivate her hypothesis, without stipulating covert mass or count light nouns. We have seen that Moltmann observes that in German, anthroponyms select d-pronouns (*der/die*) and support plural anaphora (*sie*), whereas toponyms select the neuter/mass w-pronoun (*was*) and reject plural anaphora (requiring sortals like *diese Städte*, 'those cities'). Under our framework, this asymmetry follows from the conceptualization of the referent. Anthroponyms are atomic individuals (Merge in D); being discrete, highly individuated entities, they naturally select individuating d-pronouns and, when coordinated, constitute discrete pluralities that license the anaphor *sie*. Toponyms, conversely, are structurally merged in N as aggregates. Because aggregates inherently possess 'weak individuality' (cf. Acquaviva 2008) and lack atomic boundaries, German syntax treats them akin to mass-like or continuous entities. Thus, they trigger the non-individuating w-pronoun (*was*). Furthermore, coordinating two unpartitioned expanses (i.e. *Berlin und München*) does not yield a discrete countable plurality of atoms; therefore, standard plural anaphora fails, and an overt sortal (*diese Städte*) becomes structurally obligatory to provide the boundaries necessary for countability.

Hence, the German asymmetry is not proof of silent light nouns, but a direct reflex of the atomic *vs.* aggregate nature of proper roots. One might wonder how to reconcile the fact that city names behave structurally as punctual, individual-like entities in Italian (undergoing Merge in D, as evidenced by their selection of the point-like preposition *a*; cf. Rizzi 1988; Folli 2008), whereas in German they behave as non-atomic aggregates (triggering the mass-like relative pronoun *was* and rejecting count plural anaphora; cf. Moltmann 2022). Far from undermining our hypothesis, this cross-linguistic split clearly illustrates the role of mapping parameters at the syntax-semantics interface (Longobardi 2008). Cognitively, a settlement represents a boundary entity: it can be conceptualized either as a dimensionless, punctual node (an atom) or as a spatially extended

territory (an aggregate). Italian parameterizes cities as atomic individuals (Merge in D). German, conversely, maps cities as spatial aggregates (Merge in N). The fact that German city names typically appear without a definite determiner — unlike other aggregate toponyms — does not indicate N-to-D movement. As demonstrated by D'hulst, Thieroff, & Meisenburg (2022), the absence of the article with German cities and countries is a superficial morphological reflex of their default neuter gender. Within a feature-geometric approach, the neuter is unmarked and structurally featureless, preventing the overt lexicalization of D unless the toponym bears a marked gender (e.g., feminine *die Schweiz*). Thus, the underlying syntax of German cities remains that of aggregates merged in N, explaining both their mass-like anaphoric behavior and their lack of overt determiners.

3. Merge in D vs. Merge in N: Individuals, Aggregates, and Plural Proper Names

The spatial and temporal asymmetries discussed in Section 2 suggest that the referential and aspectual properties of proper names cannot be relegated to covert classifiers. Rather, they structurally depend on the syntactic position where the proper root is initially merged. In the current theoretical literature, the syntactic status of proper names is highly debated. Predicativism (see e.g., Matushansky 2006, 2008) assumes a uniform analysis where all proper names are predicates (count nouns) merged in N, requiring an overt or covert definite determiner ($\emptyset$ *the*) to achieve referentiality. Conversely, *orthodox* Referentialism posits that names are rigid designators lacking predicative structure. Recent proposals have attempted to reconcile this ‘impasse’ through polysemy or polyreferentialism (cf. Delgado 2019; Kijania-Placek 2023, among others), suggesting that the lexicon provides an underspecified root that can activate different senses (referential, predicative, deferred) depending on the morphosyntactic environment.

Building on this latter intuition and on the standard N-to-D movement framework (Longobardi 1994, 2005, 2008; cf. also Anderson 2004, 2007), we have proposed that proper roots

are inherently conceptualized as either punctual individuals (*atoms*) or sets (*aggregates*), and this specification dictates their syntactic derivation. If a proper root denotes a punctual individual (e.g., standard anthroponyms or cities, in Italian), it is merged directly in D (or raises to D), functioning akin to a demonstrative and rejecting determiners. If, however, the root is conceptualized as a set or a collection of weakly differentiated parts (e.g., countries, plural names, and names of groups), it is merged in N just like a common noun, lacking intrinsic referential autonomy and obligatorily requiring a functional D-layer. We think that the reality of this structural dichotomy is confirmed by the behavior of *pluralized* proper names and names of groups, which are the topic of the present section.

First, if the mass/count distinction were rigidly determined by a light noun like PLACE — which is inherently noncount/mass according to Moltmann (2022) — we would not expect toponyms to display genuine plural syntax. However, certain toponyms consistently behave as morphosyntactic collectives/aggregates. In Italian, plural toponyms such as *I Caraibi* ('The Caribbean') or *Le Ande* ('The Alps') not only require a plural definite determiner, but also trigger obligatory plural agreement on the finite verb. Consider the examples in (11).

- (11) a. I Caraibi hanno /*ha un clima tropicale.
 DEF.M.PL Caribbean have.3PL/ *has a climate tropical
 'The Caribbean has a tropical climate.'
- b. *I paesi Caraibi hanno un clima tropicale.
 DEF.M.PL countries Caribbean have a climate tropical
 'The Caribbean countries have a tropical climate.'

As shown in (11b), this plural agreement cannot be attributed to the silent presence of a sortal noun like *paesi* ('countries'), because combining such a sortal directly with the toponym results in

ungrammaticality. The countability and aggregate semantics of these pluralized proper names stem from a specific syntactic construct (Merge in N) rather than a covert light noun.³

Further evidence that aggregate proper names are merged in N comes from the syntax of names of musical bands. Again, band names denote a collection of individuals with a set reference, characterized by "weak individuality" (Acquaviva 2008, cf. also Acquaviva 2025).⁴ In Italian, they consistently require a plural definite determiner and trigger plural agreement on the finite verb, entirely overriding the morphosyntactic and morphological features of their lexical root.⁵ Consider the examples in (12).

- (12) a. I Modern English sono diventati famosi con
the.M.PL Modern English are become.M.PL famous.M.PL with
I *Melt with You*.
I Melt with You
‘Modern English became famous with *I Melt with You*.’
- b. I Dead Can Dance sono stati fondati a
the.M.PL Dead Can Dance are been.M.PL founded.M.PL in
Melbourne nel 1981.
Melbourne in.the 1981
‘Dead Can Dance were founded in Melbourne in 1981.’
- c. I M/A/R/R/S sono stati i pionieri
the.M.PL M/A/R/R/S are been.M.PL the.M.PL pioneers
della musica house britannica.
of.the music house British
‘M/A/R/R/S were the pioneers of British house music.’
- d. Le Breeders sono state fondate da Kim Deal.
the.F.PL Breeders are been.F.PL founded.F.PL by Kim Deal

‘The Breeders were founded by Kim Deal.’

Within the recent literature, aggregate proper names of the type in (12) pose a challenge to the Predicativist pursuit of semantic uniformity. Standard versions of Predicativism (see Fara 2015) rely on the *Being-Called Condition* (BCC), which dictates that a proper name uniformly functions as a predicate true of the entities that are "called N". However, Rami (2014), Jeshion (2015), and Delgado (2019, 2022) have extensively documented that many systematic uses of proper names resist the BCC. These "Problematic Examples" frequently force Predicativists to either dismiss such uses as non-literal or to stipulate complex, *ad-hoc* derivation rules. Names of pluralities and aggregates lie precisely at the center of this long-standing debate. Matushansky (2015), for instance, explicitly groups the names of musical bands (e.g., the Beatles, the Rolling Stones) together with familial dynasties (the Campbells) and mountain ranges (the Alps), classifying them all as "names of plural individuals".⁶

We argue that the puzzling behavior of these aggregates is not an anomaly to be fixed by *ad-hoc* pragmatic coercion or semantic type-shifting. Rather, aggregate proper names (such, precisely, as musical bands) are inherently conceptualized as aggregates, extracted from the lexicon with plurality/aggregate features already built in (Chierchia 1998), and consequently undergo Merge directly in N. In (12a), an English phrase composed of an adjective and a singular noun (*Modern English*) is forced to take a plural masculine determiner (*i*) and trigger plural agreement. In (12b), the proper root is composed of a full clause (*Dead Can Dance*) taking the plural definite determiner; and in (12c), the same state of affairs is shown for an acronym (*M/A/R/R/S*). The fact that the plural aggregate feature entirely overrides the internal syntactic structure of a full sentence like *Dead Can Dance* proves that the external syntax is blind to the internal make-up of the proper root, reacting solely to its aggregate nature. Even more strikingly, the syntax is sensitive to the biological gender of the aggregate's members: an all-female band like *The Breeders* (12d)

obligatorily takes the feminine plural determiner *le*, overriding any default masculine assignment for foreign loanwords.⁷

If a silent light noun like PERSON were responsible for the syntax of human-denoting proper names, we would be forced to assume that PERSON arbitrarily switches its inherent features from "individual count" to "collective/aggregate mass". This is stipulative. The data in (12) naturally follow from our model, aligning with Chierchia's (1998) analysis of plurality and kinds: aggregate proper names are roots extracted from the lexicon with plurality already built in, and consequently merged in N. Because they are merged in N, they require a functional D-layer to serve as referential arguments, and the obligatory plural determiner simply reflects the syntactic realization of the aggregate feature.

As already pointed out, within the recent literature, the capacity of a proper name to modify a common noun (e.g., *The Hudson River*, or *the incomparable Maria Callas*) has been widely employed by Predicativists to argue that proper names possess internal structure and are underlyingly predicates (Matushansky 2008). Referentialists have generally struggled with these "complex names", often relying on mechanisms of deferred interpretation or ad-hoc type-shifting rules.

The aggregate nature of band names provides crucial insights into how these complex proper names are formed. When a group proper name is preceded by an overt sortal noun (e.g. *gruppo*, 'band/group'), the proper name cannot be juxtaposed as a simple compound. Consider the data in (13).

- (13) a. *Il gruppo My Bloody Valentine si è formato a Dublino nel 1983.
 the.M.SG band My Bloody Valentine REFL is formed in Dublin in 1983
- b. Il gruppo dei My Bloody Valentine si è formato a Dublino...
 the.M.SG band of.the My Bloody Valentine REFL is formed in Dublin
 'My Bloody Valentine was formed in Dublin in 1983.'

As shown in (13), the only grammatical configuration in Italian is to introduce the aggregate proper name as a prepositional modifier headed by the preposition *di* ('of'), which must incorporate the plural definite determiner (*dei*). The fact that the plural determiner is obligatorily retained inside the prepositional phrase (*de-i My Bloody Valentine*) confirms that the proper name projects its own full DP structure (merged in N) rather than acting as a bare modifying root.⁸

To account for structures like *gruppo dei My Bloody Valentine* or *Lago di Garda* without reducing proper names to regular common nouns, we must identify the exact structural primitives that the grammar utilizes to link a rigid designator to a sortal classifier. As we will demonstrate in the next Section, these structures exploit the elementary relators used for prepositional compounds and material composition (cf. Franco & Manzini 2017; Rugna & Franco 2022).

4. Bridging Onomastics and Material Constitution: Relators, Mereological Product, and Complex Names

4.1 The Puzzle of Proper Name Modification

As we have already pointed out, in current debates on the semantics of proper names, one major issue is whether names can combine with sortal nouns and modifiers. Predicativist approaches often cite expressions such as *the incomparable Maria Callas*, *the young Richard*, and *the Hudson River* as evidence that proper names are not basic referential expressions, but predicates with internal structure. As Matushansky (2008) and Fara (2015) argue, if proper names can take restrictive modifiers or determiners, they should be analyzed as count nouns with a meaning like “entity called N.” Referentialist approaches have traditionally had difficulty accounting for these constructions. — often termed “Complex Names” — resorting to mechanisms of deferred interpretation or *ad-hoc* pragmatic coercion.⁹ We have shown that Moltmann (2022), considering geographical complex names (“Type 2” names, such as rivers and mountains), attempts to circumvent the problem by

treating complex names of *the Hudson River* type as standard, lexically listed compounds headed by an overt or silent sortal.

However, cross-linguistic evidence demonstrates that complex toponyms are not simple juxtaposition compounds. In Italian, a sortal cannot directly merge with a place name; it often requires an adpositional relator, as shown in (14).

- (14) Il lago *(di) Garda
 the.M.SG lake of Garda
 ‘Lake Garda’

To explain structures like (14) without accepting the Predicativist view that *Garda* is a common count noun, we need to identify the precise structural elements that grammar uses to connect a rigid designator with a sortal classifier. We argue that this connection is not a special onomastic exception. Rather, natural languages use the same relational primitives that also encode material constitution.

4.2 The ‘Mereosyntax’ of Materials and the Dual Nature of *di*

To understand the syntax of complex names, we must first examine how the grammar encodes substances and materials. When natural languages express material constitution, they systematically map the formal ontological distinction between parts and wholes onto the syntax via a restricted set of elementary relators.

Adopting the framework developed by Manzini and colleagues (Manzini & Savoia 2011; Franco & Manzini 2017; Rugna & Franco 2022), we assume that prepositions are not semantically vacuous case-assigners, but spell out elementary mereological predicates. In Italian, material complements are structurally split into two categories: part-materials and whole-materials.¹⁰

Part-materials denote that a substance constitutes only a subpart or a surface addition to the host entity. Syntactically, they are typically introduced by the preposition *con* ('with'). Crucially, part-materials allow full DP structures, freely admitting determiners and quantifiers. Following Franco & Manzini (2017), the preposition *con* lexicalizes a reverse inclusion relator, notated as $(\supseteq)$. This structurally encodes that the material (e.g., *the bricks*) is a subpart included within the broader zonal domain of the host (e.g., *the house*). The morphosyntactic structure for a part-material is roughly represented in (15).

(15) Part-Material (Reverse Inclusion)

- a. La casa con i mattoni rossi.
‘The house with the red bricks.’
- b. $[_{DP} \text{ la casa } [_{PP(\supseteq)} \text{ con } [_{DP} \text{ i mattoni rossi}]]]$

Conversely, whole-materials denote that the entity is entirely constituted by the substance, without conceptual partitioning. Syntactically, whole-materials are introduced by the preposition *di* ('of'). Unlike part-materials, whole-materials exhibit an impoverished morphosyntactic structure that bans determiners and quantifiers.

Building on Rugna & Franco (2022) and formal mereology (Varzi 2016),¹¹ we propose that the preposition *di* in these specific configurations lexicalizes a mereological product or fusion (notated as $\otimes$).¹² The mereological product fuses the properties of the material and the host into a single, unpartitioned zonal domain (cf. Belvin & Den Dikken 1997). The impossibility of projecting a D-layer in whole-materials naturally follows from this architecture: the material root is merged directly under the fusion relator as a bare NP, structurally fusing with the head noun rather than acting as an independent referential argument. The structure for a whole-material is represented in (16).

(16) Whole-Material (Mereological Product/Fusion)

- a. Una statua di/*del marmo.
'A marble statue.'
- b. $[_{DP} \text{una statua } [_{PP(\otimes)} \text{di } [_{NP} \text{marmo}]]]$

This structural economy (bare NP *vs.* full DP) correlates directly with the interpretive rigidity of whole materials, which demand exhaustive composition, as opposed to the structural looseness of part-materials.

4.3 Compound Names as Mereological Products ($\otimes$)

We can now apply this architecture directly to *onomastics* to solve the puzzle of complex names.

We propose that the syntax of complex proper names is isomorphic to the syntax of whole-materials: the preposition *di* spells out the exact same mereological product relator ($\otimes$). However, the syntax of the modified proper name strictly depends on the fundamental cognitive dichotomy proposed in Section 3: whether the proper root denotes an atomic individual or an aggregate.

When a sortal is modified by a proper name conceptualized as a punctual, atomic individual (e.g., a city or a single geographical feature), the proper name behaves exactly like a 'bare' material root.

Because individual names bypass the N-layer and possess intrinsic referentiality (Merge in D), they do not project an overt determiner. Consequently, they merge directly under the fusion relator as a bare NP, intersecting their topological space with the sortal, as shown in (17).

(17) Proper Name Modification (Individual)

- a. Il lago di Garda.
'Lake Garda.'
- b. $[_{DP} \text{il lago } [_{PP(\otimes)} \text{di } [_{NP} \text{Garda}]]]$

The denoted geographical entity is the mereotopological fusion between the categorical properties of the sortal (*lago*) and the rigid designatory properties of the proper root (*Garda*).¹³

This structural mechanism explicitly captures the contrasting behavior of aggregate proper names (such as the names of musical bands discussed in Section 3). When a sortal noun like *gruppo/band* ('group/band') is modified by an aggregate proper name (e.g., *My Bloody Valentine*), the proper root cannot merge as a bare NP under the preposition *di*. Because aggregates possess internal mereological complexity (Merge in N), they lack intrinsic referential autonomy and obligatorily require a functional D-layer (the plural article) to serve as referential arguments.

Consequently, the aggregate proper name is forced to project its full DP structure — bringing its obligatory plural determiner into the prepositional phrase and generating the articulated *dei/degli* (*di* plus *i/gli*), as illustrated in (18).

(18) Aggregate Proper Name Modification

a. La band dei/*di My Bloody Valentine/degli/*di Stereolab.

‘The My Bloody Valentine/Stereolab band.’

b. [_{DP} la band [_{PP(⊗)} di [_{DP} i My Bloody Valentine]]]

The contrast between [_{PP} di [_{NP} *Garda*]] and [_{PP} di [_{DP} i My Bloody Valentine]] is theoretically crucial. It proves that the presence or absence of the determiner in complex names is not dictated by idiosyncratic metalinguistic rules or by the sortal noun itself. Rather, it is the direct morphosyntactic reflex of the underlying mereological nature of the proper root: atomic individuals fuse as bare NPs, while conceptual aggregates obligatorily project a full DP layer even when acting as modifiers.¹⁴

4.4 Morphological Evidence: Fusion vs. Inclusive Suffixes

Striking independent evidence for this unified mereosyntactic approach comes from the domain of adjectival derivation. As noted by Franco & Lorusso (2022) (cf. Marchis Moreno 2018, Alexiadou

& Stavroi 2011), relational interpretations of materials and substances can be semantically and syntactically typed via different mereological relations. Natural languages employ specific morphological suffixes that mirror the relational value of the corresponding prepositions.

In Italian, adjectives describing materials, substances, and environments display a sharp morphological split that reflects a broad spectrum of mereological relators. First, specific suffixes encode inclusion/part whole ($\subseteq$). Franco & Lorusso (2022b) note that adjectives like *marina* ('marine') in *acqua marina* ('sea water') instantiate a part-whole relation where the head noun (*acqua*) is conceptually included as a part of a greater contextual entity (*mare*). This adjectival derivation structurally mirrors the preposition *di* in its inclusive/possessive use, as illustrated by the bracketed representations in (19).

(19) Inclusion ($\subseteq$)

- a. [NP acqua [PP [P di ($\subseteq$) [NP mare]]]]
 - b. [NP acqua [[√mar] [-in ($\subseteq$) [infl -a]]]]
- ‘Sea water’

Second, other suffixes strictly encode reverse inclusion ($\supseteq$), of the type illustrated in (15) above.

Suffixes like *-oso* and *-ato* mirror the preposition *con* ('with'). For instance, a *terreno acquoso* ('watery soil') or a *vaso argentato* ('silver-plated vase') implies that the material is only located on the surface or distributed across the host entity, without exhaustively constituting it. The structural isomorphism between the prepositional and the adjectival reverse inclusion is illustrated in (20).

(20) Reverse Inclusion ($\supseteq$)

- a. [NP terreno [PP [P con ($\supseteq$) [NP acqua]]]]
- b. [NP terreno [[√acqu] [-os ($\supseteq$) [infl -o]]]]

Finally, suffixes such as *-eo* and *-ico* encode mereological intersection or fusion ($\otimes$), directly mirroring the whole-material use of the preposition *di* ('of'). Adjectives like *argenteo* ('silvery' / 'made of silver') or *aureo* ('golden' / 'made of gold') denote an entity entirely pervaded by the material. Similarly, the highly productive suffix *-ico* serves as a fusion relator when applied to material bases: expressions like *struttura metallica* ('metallic structure', i.e., a structure made entirely of metal) indicate that the object is exhaustively constituted by the substance, without conceptual partitioning.¹⁵

The grammar of proper names replicates this morphological pattern when proper roots are converted into ethnic or geographical adjectives modifying a sortal. In Italian, complex names of seas and oceans systematically utilize fusion-encoding suffixes to link the proper root to the sortal head (*mare*, 'sea', or *oceano*, 'ocean').

- (21) a. Il mar Mediterran-eo / Eg-eo
the sea Mediterranean / Aegean
'The Mediterranean / Aegean Sea'
- b. Il mar Balt-ico / Adriat-ico
the sea Baltic / Adriatic
'The Baltic / Adriatic Sea'

In (21), the suffixes *-eo* and *-ico* attached to the proper root perform the exact same morphosyntactic function as they do in *argenteo* (silver) or *metallico* (metallic): they realize an intersection relator ($\otimes$). Consequently, a prepositional construct like *Mare di Alboran* and an adjectival construct like *Mar Egeo* are structurally (i.e mereologically) identical. The rigid designator provides the defining property that pervades the entire geographical entity, generating an intersected, unpartitioned topological domain.

This structural architecture provides a possible interpretation to the puzzle of complex names. When a proper name modifies a sortal, it does not need to be coerced into a predicative count noun (e.g., "the sea called Baltic", as Predicativism would enforce). Instead, it maintains its object-referring or fixed set-denoting status, whose topological space is simply fused with that of the sortal via the $\otimes$ relator. The presence of a relator in Romance complex toponyms shows that the syntax is not dealing with a metalinguistic predicate (a count noun) modifying another noun. Instead, the syntax is performing a *mereo-topological* fusion: the fixed spatial domain of the proper root is intersected directly with the categorical domain of the sortal.

5. The Double Life of Proper Names

The morphosyntactic evidence discussed so far has some implications for the ongoing debate regarding the semantic status of proper names. Both Referentialism and Predicativism face challenges when confronted with the full spectrum of onomastic data. Referentialism struggles to explain how names can be systematically pluralized, modified, or combined with sortals without losing their referential integrity. Predicativism, by contrast, assigns proper names an unusual and highly stipulative syntactic distribution, without explaining why, across languages, they behave so differently from ordinary count nouns. As Delgado (2019) and Jeshion (2015, 2017) point out, the pursuit of uniformity via a null determiner is a theoretical artifact that obscures the profound morphosyntactic asymmetries between names and nouns.

To overcome this impasse, several scholars have recently proposed “polysemous” or “polyreferential” approaches (Leckie 2013; Jeshion 2015; Delgado 2019; Kijania-Placek 2023). On these views, the lexicon provides an underspecified root that can yield different interpretations—referential, predicative, or deferred—depending on the context. A systematic version of this idea is the *Allosemy View* proposed by Saab and Lo Guercio (2019). They argue that grammatical categories and their meanings are not primitive, but emerge from the way syntax combines

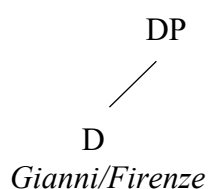
functional material with lexical roots. On this view, lexical roots do not have identifiable meanings before entering the syntactic derivation: the abstract lexicon does not contain pre-syntactic “names,” “verbs,” or “nouns.” Our analysis provides the precise syntactic machinery needed to formalize this insight. The so-called “Being Called Condition” of the Predicativists is neither a syntactic nor a semantic primitive. Instead, the specific referential or predicative life of a name is structurally determined by its conceptual ‘mereology’ (atoms vs. aggregates) and its subsequent point of attachment in the nominal spine.

5.2 The Syntactic Structures of Properhood in Italian

By dispensing with silent light nouns (Moltmann 2022) and covert definite articles (Fara 2015), we can map the different “flavors” of proper names onto a highly restricted set of possible syntactic structures. The fundamental distinction lies between Merge in D and Merge in N (eventually mediated by elementary relational primitives).

Let’s illustrate the typology, starting from Atoms/Punctual Individuals. When an onomastic root is conceptualized as a punctual individual (an atom), it bypasses the N-layer. It is merged directly in D,¹⁶ functioning as a fully referential, demonstrative-like element. Because D is explicitly filled by the proper root, determiners are strictly banned. This structure applies to standard anthroponyms and punctual toponyms (e.g., cities like *Firenze* or *Milano* in Italian).¹⁷

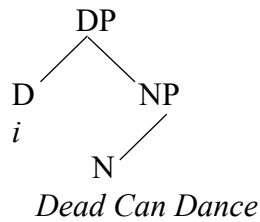
(22) Individual Proper Name (Referential)



When an onomastic root is conceptualized as an aggregate or a set characterized by weak individuality (e.g., plural names, musical bands like *I Dead Can Dance*, or extensive geographical

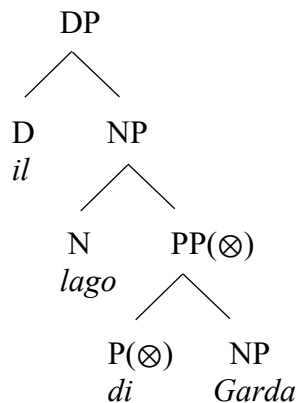
areas like *Il Portogallo*), it enters the derivation exactly like a common noun. It merges in N and obligatorily requires a functional D-layer to serve as an argument.¹⁸ The overt determiner is not a predicativist "repair" mechanism, but the standard functional head closing a set-denoting projection.

(23) Aggregate Proper Name



Finally, when an onomastic root modifies an overt sortal noun (e.g., *il lago di Garda*, *il mar Egeo*), the grammar does not resort to semantic coercion, nor does it transform the name into a metalinguistic predicate ("the lake called Garda"). Instead, it exploits mereotopological morphosyntax akin to material constitution. The preposition *di* (or adjectival suffixes like *-eo*) spells out the elementary fusion relator ($\otimes$).¹⁹ The proper root merges as an NP under the relator, intersecting its referential properties with the sortal head to form a single, unpartitioned zonal domain.

(24) Complex Proper Name (Intersection/Whole-Material)



In conclusion, the "double life" of proper names — their oscillation between singularity and generality, and their capacity to modify other nouns — does not necessitate abandoning Referentialism for Predicativism. It simply requires recognizing that natural languages generate reference and modification through a limited set of structural positions (D vs. N) and elementary relators. By proposing this mereosyntactic architecture, our framework avoids the problems faced by both strict Referentialism and uniform Predicativism. We do not assume that onomastic roots are always associated with a single semantic type, whether as pure referential tags or as count nouns. Rather, in line with recent work on category shifts and polyreferentialism (e.g. Delgado 2019), we argue that the referential or predicative behavior of a proper name is derived from its structural position and from its mereological construal, that is, whether it is conceptualized as an atom or as an aggregate.

6. The Universality of the Individual vs. Set Dichotomy: A Cross-Linguistic Perspective

From a UG perspective, if strict Predicativism were correct, proper names should systematically mirror the morphosyntactic behavior of common count nouns across all languages. Conversely, if Moltmann's (2022) Light Noun Hypothesis were on the right track, the syntax of proper names should be uniformly governed by a universal set of covert classifiers (e.g., PERSON, PLACE), producing homogeneous syntactic classes.

Typological evidence challenges both assumptions. Cross-linguistic data reveal that natural languages grammaticalize the exact individual *vs.* spatial aggregate dichotomy that we have proposed for Italian.

6.1 Differential Object Marking (DOM) and the Limits of Covert Classifiers

Evidence supporting the structural divide between atomic individuals (Merge in D) and aggregates/spatial expanses (Merge in N) comes from the domain of Differential Object Marking

(DOM). As is well known from typological studies (Bossong 1985; Aissen 2003), DOM is a phenomenon whereby certain direct objects are overtly case-marked depending on their relative prominence on the Animacy and Definiteness/Referentiality scales.

Typological research demonstrates that proper names do not occupy a single, uniform slot on the Animacy Hierarchy. While standard anthroponyms rank extremely high—typically occupying a prominent position between personal pronouns and human common nouns (Dixon 1979; Helmbrecht 2014; Caro Reina & Helmbrecht 2022) — toponyms rank significantly lower. This scalar divergence is directly reflected in DOM assignment. Within Romance languages, for instance, the adposition *a* ('to/at') frequently acts as the DOM marker. Consider the contrast between anthroponyms and toponyms in standard Spanish, as illustrated in (25).

- (25) a. Visité (a) María.
 visited.1SG DOM Maria
 ‘I visited Maria.’
- b. Visité *(a) Madrid.
 visited.1SG DOM Madrid
 ‘I visited Madrid.’

Personal names (anthroponyms) occupy a high position on the referentiality hierarchy and systematically trigger obligatory DOM (25a). Conversely, toponyms (like cities or countries) systematically reject it in standard contexts (25b), patterning instead with inanimate common nouns. As noted by Laca (2006) and Haspelmath (2019), this asymmetry is rooted in the diachrony of Romance: in Old Spanish DOM was strictly obligatory for personal names, but remained optional or entirely absent for place names.

If strict Predicativism were correct in assuming that all proper names are underlyingly common count nouns embedded under a (covert) definite determiner, the syntactic split in (25) would remain

an unexplained anomaly. Why should one "count noun" obligatorily trigger DOM while another does not, if their syntactic architecture is identical? Similarly, Moltmann's (2022) Light Noun Hypothesis does not provide an adequate explanation in this case. If the syntax of these names simply relied on the covert classifiers PERSON and PLACE, one would be forced to arbitrarily stipulate that DOM is sensitive to lexical classifiers. This merely restates the problem without offering a structural explanation for *why* PLACE behaves like a lower-ranked common noun while PERSON behaves like a pronoun/high referential individual. Our framework provides a structural account for this asymmetry. Following Manzini and Franco (2016), we assume that DOM markers (like the Romance *a*) lexicalize an elementary inclusion relator ($\sqsubseteq$), which is structurally required to integrate highly individuated, autonomous arguments into the event structure (as 'possessors' of the event). Because personal names are conceptualized as atomic individuals devoid of internal mereological complexity, they possess intrinsic referentiality and undergo Merge directly in D. It is precisely this D-level syntactic atomicity that requires the specialized $\sqsubseteq$ relational embedding (DOM). Toponyms, on the other hand, exhibit a lower degree of onomastic prototypicality. Because macro-places are cognitively parsed as mereo-topological expanses, boundaries, or sets of locations rather than atomic individuals, they lack intrinsic D-level referentiality.²⁰ Consequently, they undergo Merge in N, requiring external functional layers to act as arguments. By merging in N, they align structurally with 'standard' common nouns,²¹ falling below the threshold that triggers the $\sqsubseteq$ relator. The syntactic variation in DOM is thus not the reflex of an idiosyncratic lexical rule or a silent classifier, but the structural consequence of the individual (*Merge in D*) versus aggregate/expanse (*Merge in N*) dichotomy.

6.2 The Individual/Set Split in the Spatial Domain: Beyond Light Nouns

Cross-linguistic studies demonstrate that natural languages systematically employ asymmetric coding for spatial roles (locative, allative, ablative) depending on the referential nature of the noun. Specifically, in a massive typological sample, Stolz et al. (2014) demonstrate that toponyms often

undergo zero-marking (the complete omission of spatial adpositions or determiners) in contexts where common nouns and personal names obligatorily require overt relational marking. Oceanic languages offer striking examples of this structural split. Consider the contrast in Maori, which utilizes the proprial article *a* for proper names and the common article *te* for common nouns. However, as described in Bauer's (1993: 109) reference grammar, this apparent ‘uniformity’ within the onomastic class breaks down in adpositional contexts. When governed by the directional preposition *ki* ('to/towards/with'), personal names obligatorily retain their proprial article. Conversely, place names must be radically zero-marked, completely dropping the article. This morphosyntactic asymmetry is a general rule of the language, as illustrated by the following sentences (drawn from Bauer 1993 and cited in Stolz & Levkovych 2022).

(26) a. Anthroponym (Obligatory article retention)

Ka aahua pukuriri a Tamahae [ki a Rewi].
TMA somewhat angry PROP Tamahae DIR PROP Rewi
‘Tamahae was somewhat angry with Rewi.’

(Bauer 1993: 92; Stolz & Levkovych 2022: 255)

b. Toponym (Obligatory article drop)

I haere atu a ia [ki Ø Ingarangi me Amerika].
TMA move away PROP 3SG DIR Ø England and America
‘He travelled to England and America.’ (Bauer 1993: 129; Stolz & Levkovych 2022: 256)

If Predicativism were correct, toponyms and anthroponyms would share the exact same underlying architecture as common count nouns. Thus, it would remain mysterious why the syntax systematically strips away the functional D-layer exclusively for geographic names in spatial contexts. Moltmann’s Light Noun Hypothesis is no more successful here. Assuming a covert PLACE classifier in (26b) and a covert PERSON classifier in (26a) amounts only to a descriptive

label. It does not provide a principled structural account of why the PLACE classifier regularly leads to morphological reduction and zero-marking, whereas the PERSON classifier does not. Our framework derives this Differential Place Marking (Haspelmath 2019) and the zero-marking of toponyms directly from the underlying structural architecture: anthroponyms are cognitively and structurally conceptualized as atomic, punctual individuals. Because an atom lacks internal mereotopological geometry, it is conceptually incompatible with functioning as a spatial enclosure. To be coerced into a spatial destination or target (e.g., "with/to Rewi"), the atomic individual (merged in D) requires robust morphosyntactic support, retaining its full proprial morphological device *a* to mediate the spatial relation.

Toponyms, conversely, denote mereotopological expanses and unpartitioned sets (Merge in N). Because they are inherently extended spatial domains, they naturally satisfy the topological requirements of the spatial event without the need for additional functional mediation. The syntax maps this conceptual expanse by bypassing the overt relational embedding at the D-layer (zero-marking), fusing the spatial domain of the verb/preposition directly with the spatial domain of the toponym.²² The Differential Place Marking phenomenon is thus a direct morphosyntactic reflex of the root's mereological nature: atoms require structural mediation to be located in space, whereas spatial sets do not.

Moreover, the toponymic split discussed in Section 2 — where punctual, individual-like cities reject determiners, while set-like countries require them — is not an Italian quirk. It reflects a possible universal cognitive (and syntactic) distinction between individuals and spatial extensions. Recent typological investigations (Haspelmath 2019; Stolz & Levkovich 2022; Winistörfer et al. 2025) provide cross-linguistic evidence that toponyms do not constitute a homogeneous grammatical class. Rather, they bifurcate according to their spatial extension and conceptual boundedness.

An empirical confirmation of this internal toponymic split comes from the Balkan linguistic area, specifically from Aromanian and Arbëreshë (cf. Manzini & Savoia 2018). As documented by

Stolz and Levkovich (2019), spatial relations in these languages are marked differentially depending strictly on whether the ground is a settlement name (an individual, point-like entity) or a country/region (an extended aggregate). In Aromanian, for instance, directional and locative relations are obligatorily zero-marked when the complement is a city or village, as in (27a). However, if the toponym denotes a country or a large region, zero-marking is strictly ungrammatical, and an overt spatial preposition is required, as in (27b).

- (27) a. [Ø Krušuva] hǝvǎja ǰǎste ávrǝ.
 Ø Kruševo air.DEF be.3SG fresh.F
 ‘In Kruševo, the air is fresh.’ (Merge in D: Individual)
- b. vrém ta-s-fúg [tu Eládǝ].
 want.1SG.IMPERF SUBORD-CONJ-go.1SG in Greece
 ‘I wanted to go to Greece.’ (Merge in N: Aggregate)

Aromanian (adapted from Gołąb 1984; Stolz & Levkovich 2019)

The exact same systematic split is found in Arbëreshë varieties (Italo-Albanian), where settlement names like *Moscherato* take zero spatial marking (Ø *Muskaritull*), while regions like *Calabria* obligatorily require the preposition *nd'* (*nd' Kallabriet*).

This cross-linguistic mapping aligns with our ‘mereosyntactic’ dichotomy. Cities and small settlements are often cognitively and structurally parsed as punctual atoms: they undergo Merge directly in D, acquiring independent referential autonomy, which allows them to bypass standard prepositional or determiner marking (zero-marking / the preposition *a* in Italian). Conversely, countries and macro-regions lack this punctual atomic status; they are mereotopological expanses (sets). Consequently, they undergo Merge in N and require an overt functional layer (a D-layer or a

full Prepositional Phrase, such as *tu* in Aromanian or *in + D* in Italian) to function as spatial arguments.

Additionally, the mereological architecture proposed in Section 4 to account for complex names (proper names modifying sortals) finds robust cross-linguistic validation. As investigated by D'hulst, Thieroff, and Meisenburg (2022), the morphosyntax of geographical complex names (e.g., river names, lake names) is parametrically split between Romance and Germanic languages.

In Romance languages, the fusion of the categorial properties of the sortal and the designatory properties of the proper root cannot be achieved via simple juxtaposition; it requires a functional relator spelling out the mereological product ($\otimes$). Thus, Italian, Spanish, and French consistently employ the adpositions *di/de* (e.g., *Lago di Garda*, *Río de la Plata*, *Golfe de Gascogne*).

Conversely, Germanic languages achieve this ‘fusion’ without adpositional relators, relying instead on morphological compounding or N-N juxtaposition, often mirroring the morphosyntax of their material compounds.²³

- (28) a. [Lake [Windermere]] / [River [Thames]] (English)
b. [Hudson [River]] / [Loch [Ness]] (English)
c. [Bodensee] / [Rhein-fluss] (German)

We argue that while the syntactic mechanism varies (adpositional relator in Romance *vs.* compounding in Germanic), the underlying operation remains identical: the proper root never acts as a standard predicative count noun taking the definite article independently. Rather, it fuses with the sortal head to define a single, intersectional zonal domain. This confirms that proper name modification is governed by mereological primitives rather than arbitrary onomastic rules.

A further proof that the projection of the D-layer depends on the aggregate or extended nature of the referent comes from the cross-linguistic morphosyntax of river names.

If Predicativism were right, the absence of the definite article with cities (e.g., *Rome*, *Paris*) and its presence with rivers (e.g., *the Thames*, *the Nile*) would simply be a matter of arbitrary, language-specific rules of pronunciation for a covert determiner. However, as demonstrated by typological research, river names behave exceptionally and uniformly across different language families: they obligatorily require the definite article in all Germanic and Romance languages that possess prenominal determiners.²⁴

This behavior challenges both Predicativism and the Light Noun Hypothesis. A river is not an atomic individual. Due to its massive *longitudinal* extension and flowing nature, a river is conceptually unbounded; it is an aggregate of moving substance that lacks the punctual, delimited borders of a city. Because river names denote unbounded spatial extensions, they are topologically prohibited from merging directly in D (unlike ‘atomic’ cities). They must undergo Merge in N. Consequently, just like the names of large countries or plural archipelagos, river names obligatorily require the projection of a functional D-layer (the definite article) to resolve their mereological unboundedness and act as referential arguments in the syntax.

In conclusion, the morphosyntactic behavior of proper names is not dictated by uniform predication rules or arbitrary covert classifiers. It is the direct structural realization of a fundamental mereotopological dichotomy. Roots conceptualized as discrete, punctual atoms merge in D, overriding common noun syntax. Roots conceptualized as sets, aggregates, or unbounded spatial extensions (like bands, nations, and rivers) merge in N, requiring functional D-support to achieve argumenthood. Cross-linguistic evidence confirms our claim: roots conceptualized as discrete, punctual individuals merge in D, driving distinct case alignment, DOM, and overriding agreement (see fn. 17). Roots conceptualized as sets, aggregates, or spatial extensions merge in N, requiring functional support (like determiners or adpositions) to achieve argumenthood.

7. Conclusion

In this paper, we have challenged both the strict Predicativist search for uniformity and the structural assumptions of the Light Noun Hypothesis. By examining the morphosyntax of proper names, mainly in Italian and with supporting typological evidence, we have shown that proper names do not form a single uniform morphosyntactic class, nor is their behavior determined by arbitrary covert classifiers such as PLACE or PERSON.

Instead, we have argued that the “double life” of proper names reflects a basic cognitive and structural distinction at the syntax-semantics interface. Onomastic roots can be conceptualized either as atomic, individual entities or as sets, aggregates, and extended mereotopological domains. This distinction determines their syntactic derivation. Names denoting individuals are merged in D, and therefore display referential autonomy similar to that of demonstratives and do not combine with determiners. By contrast, names denoting aggregates, such as country names, plural toponyms, and names of musical groups, are merged in N. Since they lack inherent referential autonomy, they require a functional D-layer in order to function as arguments.

Crucially, this structural architecture also offers a clear solution to the long-standing problem of Complex Names, that is, cases in which proper names modify sortal nouns. Predicativists have often treated such constructions as evidence that all proper names are underlyingly predicates. We have argued instead that natural languages solve this problem by using the same relational primitives that are used to express material constitution. Specifically, we have proposed that the Italian preposition *di* ‘of’ in complex toponyms such as *Lago di Garda*, as well as certain adjectival suffixes such as *-eo* in *Mar Egeo*, realize a mereological product or fusion relator ($\otimes$). This relator combines the descriptive content of the sortal with the designating content of the proper root into a single, non-partitioned zonal domain, without forcing the proper name to become an ordinary predicative count noun.

By bringing together onomastics, generative syntax, and mereology, we hope to have offered a robust framework for understanding the grammatical nature of proper names.

References

- Acquaviva, Paolo. 2008. *Lexical plurals*. Oxford: Oxford University Press.
- Acquaviva, Paolo. 2025. Names and naming verbs. *Annali di Ca' Foscari. Serie occidentale* 59. 9–30.
- Adger, David. 2025. *Mereological syntax: Phrase structure, cyclicity, and islands*. Cambridge, MA: MIT Press.
- Aissen, Judith. 2003. Differential object marking: Iconicity vs. economy. *Natural Language and Linguistic Theory* 21(3). 435–483.
- Alexiadou, Artemis and Melita Stavrou. 2011. “Ethnic Adjectives as Pseudo-Adjectives: A Case Study on Syntax–Morphology Interaction and the Structure of DP.” *Studia Linguistica* 65(2): 117–152.
- Anderson, John M. 2004. On the grammatical status of names. *Language* 80(3). 435–474.
- Anderson, John M. 2007. *The grammar of names*. Oxford: Oxford University Press.
- Asher, Nicholas, and James Pustejovsky. 2005. Word meaning and commonsense metaphysics. Ms.
- Bauer, Winifred. 1993. *Maori*. London: Routledge.
- Bach, Kent. 2015. The predicate view of proper names. *Philosophy Compass* 10(11). 772–784.
- Bossong, Georg. 1985. *Empirische Universalienforschung: Differentielle Objektmarkierung in den neuiranischen Sprachen*. Tübingen: Narr.
- Burge, Tyler. 1973. Reference and proper names. *The Journal of Philosophy* 70(14). 425–439.
- Burnett, Heather. 2006. On the status of proper names in the grammar. In *Proceedings of the 2006 annual conference of the Canadian Linguistic Association*, ed. by Claire Gurski & Milica Radisic, 1–15. Boston, MA: Cascadilla Press.
- Caro Reina, Javier and Johannes Helmbrecht, eds. 2022. *Proper names versus common nouns: Morphosyntactic contrasts in the languages of the world*. Berlin: De Gruyter Mouton.

- Casati, Roberto & Achille C. Varzi. 1999. *Parts and places: The structures of spatial representation*. Cambridge, MA: MIT Press.
- Chierchia, Gennaro. 1998. Reference to kinds across languages. *Natural Language Semantics* 6(4). 339–405.
- Cinque, Guglielmo. 2023. On the Merge position of additive and associative plurals. In *From sounds to structures: Beyond the veil of Maya*, ed. by Roberto Petrosino, Pietro Cerrone & Harry van der Hulst, 499–513. Berlin: De Gruyter Mouton.
- Comrie, Bernard. 1981. Ergativity and grammatical relations in Kalaw Lagaw Ya (Saibai dialect). *Australian Journal of Linguistics* 1. 1–42.
- De Rijk, Rudolf P. G. 2008. *Standard Basque: A progressive grammar*. Cambridge, MA: MIT Press.
- Delgado, Laura. 2019. Between singularity and generality: The semantic life of proper names. *Linguistics and Philosophy* 42(4). 381–417.
- Delgado, Laura. 2022. Names vs nouns. *Philosophical Studies* 179(11). 3233–3258.
- D'hulst, Yves, Rolf Thieroff & Trudel Meisenburg. 2022. River names. In *Proper names versus common nouns*, ed. by Javier Caro Reina & Johannes Helmbrecht, 93–120. Berlin/Boston: De Gruyter Mouton.
- Dixon, Robert M. W. 1979. Ergativity. *Language* 55. 59–138.
- Fara, Delia Graff. 2015. Names are predicates. *The Philosophical Review* 124(1). 59–117.
- Folli, Raffaella. 2008. Complex PPs in Italian. In *Syntax and Semantics of Spatial P*, ed. by Anna Asbury, Jakub Dotlačil, Berit Gehrke, and Rick Nouwen, 197–220. Amsterdam: John Benjamins.
- Franco, Ludovico & M. Rita Manzini. 2017. Instrumental prepositions and case: Contexts of occurrence and alternations with datives. *Glossa: a journal of general linguistics* 2(1). 1–47.

- Franco, Ludovico & Paolo Lorusso. 2019. The expression of proper locations and beyond: The motion-to and state-in Italian spatial prepositions. In *Proceedings of NELS 49*, vol. 1, 279–290. Amherst, MA: GLSA.
- Franco, Ludovico & Paolo Lorusso. 2022a. On the interpretation of proper temporal adverbs in Italian: The role of P and D. *Isogloss* 8(2)/19. 1–18.
- Franco, Ludovico & Paolo Lorusso. 2022. Derivational relators in Italian. *Languages* 7:130.
- Franco, Ludovico & Giuseppe Rugna. 2024. The morphosyntax of proper names: Individuals, sets and the light noun hypothesis. *Quaderni di Linguistica e Studi Orientali* 10. 9–27.
- Haspelmath, Martin. 2019. Differential place marking. *Linguistics* 57(1). 1–28.
- Helmbrecht, Johannes. 2014. On the form and function of proper names: A typological perspective. Ms., University of Regensburg.
- Jackendoff, Ray. 1991. Parts and boundaries. *Cognition* 41(1–3). 9–45.
- Jeshion, Robin. 2015. Referentialism and predicativism about proper names. *Erkenntnis* 80(Suppl. 2). 363–404.
- Jeshion, Robin. 2017. The problem for the-predicativism. *The Philosophical Review* 126(2). 219–240.
- Kayne, Richard S. 2005. *Movement and silence*. Oxford: Oxford University Press.
- Kayne, Richard S. 2010. *Comparisons and contrasts*. Oxford: Oxford University Press.
- Kripke, Saul A. 1980. *Naming and necessity*. Cambridge, MA: Harvard University Press.
- Laca, Brenda. 2006. El objeto directo preposicional en español. In *Sintaxis histórica de la lengua española: Primera parte: La frase verbal*, ed. by Concepción Company Company, 423–475. Mexico City: Universidad Nacional Autónoma de México.
- Langendonck, van Willy. 2007. *Theory and typology of proper names*. Berlin: Mouton De Gruyter.
- Leckie, Gail. 2013. The double life of names. *Philosophical Studies* 165(3). 1139–1160.

- Link, Godehard. 1983. The logical analysis of plurals and mass terms: A lattice-theoretical approach. In *Meaning, use, and interpretation of language*, ed. by R. Bäuerle, C. Schwarze & A. von Stechow, 303–323. Berlin: De Gruyter.
- Longobardi, Giuseppe. 1994. Reference and proper names: A theory of N-movement in syntax and logical form. *Linguistic Inquiry* 25(4). 609–665.
- Longobardi, Giuseppe. 2005. Toward a unified grammar of reference. *Zeitschrift für Sprachwissenschaft* 24(1). 5–44.
- Longobardi, Giuseppe. 2008. Reference to individuals, person, and the variety of mapping parameters. In *Essays on nominal determination: From morphology to discourse management*, ed. by Henrik Høeg Müller & Alex Klinge, 189–211. Amsterdam: John Benjamins.
- Manzini, M. Rita & Leonardo M. Savoia. 2011. Reducing 'case' to denotational primitives. *Linguistic Variation* 11(1). 76–120.
- Manzini, M. Rita, & Leonardo M. Savoia. 2018. *The Morphosyntax of Albanian and Aromanian Varieties. Case, Agreement, Complementation*. Berlin: Mouton de Gruyter.
- Manzini, M. Rita, & Ludovico Franco. 2016. Goal and DOM datives. *Natural Language and Linguistic Theory* 34(1). 197–240.
- Marchis Moreno, Mihaela. 2018. *Relational adjectives in Romance and English: Mismatches at interfaces*. Cambridge: Cambridge University Press.
- Marinis, Michail I. 2025. On the morphology of toponyms: What (Greek) inflectional paradigms can teach us. *Transactions of the Philological Society* 123(1). 77–96.
- Matushansky, Ora. 2008. On the linguistic complexity of proper names. *Linguistics and Philosophy* 31(5). 573–627.
- Matushansky, Ora. 2015. The other Francis Bacon: On non-bare proper names. *Erkenntnis* 80(Suppl. 2). 335–362.

- Mauri, Caterina & Andrea Sansò. 2019. Nouns & co.: Converging evidence in the analysis of associative plurals. *Language Typology and Universals* 72(4). 603–626.
- Moltmann, Friederike. 2022. Names, light nouns, and countability. *Linguistic Inquiry* 54(1). 117–146.
- Ortmann, Albert. 2014. Definite article asymmetries and concept types: Semantic and pragmatic uniqueness. In *Frames and concept types: Applications in language and philosophy*, ed. Thomas Gamerschlag, Doris Gerland, Rainer Osswald, and Wiebke Petersen, 293–321. Cham: Springer.
- Rabern, Brian. 2015. Descriptions which have grown capital letters. *Mind & Language* 30(3). 292–319.
- Rami, Dolf. 2014. The use-conditional indexical conception of proper names. *Philosophical Studies* 168(1). 119–150.
- Rami, Dolf. 2015. The multiple uses of proper nouns. *Erkenntnis* 80(Suppl. 2). 405–432.
- Rizzi, Luigi. 1988. Il sintagma preposizionale. In *Grande grammatica di consultazione*, vol. 1, ed. by Lorenzo Renzi. Bologna: Il Mulino.
- Rugna, Giuseppe & Ludovico Franco. 2022. Prepositions as relators in Italian prepositional compounds. *Isogloss* 8(5). 1–21.
- Salaberri, Iker. 2022. D-marking on Basque personal names from a synchronic and diachronic perspective. In *Proper names versus common nouns: Morphosyntactic contrasts in the languages of the world*, ed. by Javier Caro Reina & Johannes Helmbrecht, 205–236. Berlin/Boston: De Gruyter Mouton.
- Schoubye, Anders J. 2017. Type-ambiguous names. *Mind* 126(503). 715–767.
- Smith, Barry. 1995. Formal ontology, common sense and cognitive science. *International Journal of Human-Computer Studies* 43. 646–667.
- Stolz, Thomas, Sander Lestrade & Christel Stolz. 2014. *The crosslinguistics of zero-marking of spatial relations*. Berlin: De Gruyter Mouton.

- Stolz, Thomas, and Nataliya Levkovych. 2022. On Special Onymic Grammar (SOG): Definiteness markers in Fijian and selected Austronesian languages. In *Proper names versus common nouns: Morphosyntactic contrasts in the languages of the world*, ed. Javier Caro Reina and Johannes Helmbrecht, 237–264. Berlin: De Gruyter Mouton.
- Supalla, Samuel J. 1990. The arbitrary name sign system in American Sign Language. *Sign Language Studies* 67. 99–126.
- Trask, R. L. 2003. The noun phrase: nouns, determiners and modifiers; pronouns and names. In *A grammar of Basque*, ed. by José Ignacio Hualde & Jon Ortiz de Urbina, 113–171. Berlin: Mouton de Gruyter.
- Van de Velde, Mark & Odette Ambouroue. 2011. The grammar of Orungu proper names. *Journal of African Languages and Linguistics* 32(1). 113–141.
- Varzi, Achille C. 2016. “Mereology.” In *The Stanford Encyclopedia of Philosophy*, ed. Edward N. Zalta.
- Winistörfer, Olivier, Anastasia Escher & Daria Konior. 2025. Lost in the Balkans: Differential place marking in the Aromanian varieties. *Journal of Contemporary Philology* 8(1). 129–143.
- Zamparelli, Roberto. 2000. *Layers in the determiner phrase*. New York: Garland Publishing.

¹ For broader theoretical defenses of the predicative status of proper names and their interaction with determiners, see also Bach (2015) and Rabern (2015).

² It is worth noting that while we adopt standard minimalist terminology (*Merge*) and Longobardi’s (1994, 2005) framework to capture this structural dichotomy, our mereological approach to proper names is intrinsically compatible with the recent program of *Mereological Syntax* (Adger 2025). If syntactic objects are built via asymmetric part-whole relationships (*Subjoin*) rather than set-theoretic *Merge*, the individual vs. aggregate distinction proposed here straightforwardly translates into atomic roots subjoining directly to D versus complex aggregates subjoining to N.

³ Furthermore, cross-linguistic research has extensively documented the phenomenon of "associative plurals", where a proper name denotes a focal referent and an associated group. Italian makes productive use of associative constructs, triggering plural agreement — e.g., *Obama e i suoi hanno..* ('Obama and his team.PL have...'). As discussed by Mauri and Sansò (2019) and Cinque (2023), the semantics of associative plurals is radically different from additive plurals: they denote a heterogeneous set centering around a focal referent.

⁴ Although we elect to use musical bands as our primary testing ground for aggregate proper names, this morphosyntactic behavior is by no means restricted to them. The obligatory projection of a D-layer to encode aggregate-reference systematically extends to family names and dynasties (e.g., *the Kennedys*, *the Romanovs*; or Italian surnames: *i Rossi*), as well as to geographical *pluralia tantum* like mountain ranges and archipelagos (e.g., *the Alps*, *the Pyrenees*, *the Hebrides*, cf. Langendonck 2007).

⁵ One might wonder why a toponym like *Portogallo* ('Portugal'), if analyzed as a set or an aggregate, exhibits singular morphology and triggers singular agreement, unlike plural aggregates such as *i Caraibi* ('The Caribbean'). It is crucial to distinguish between morphological number and mereotopological complexity. As argued in Franco and Rugna (2024), country names encode a set of referents (e.g., cities, mountains, regions) within a single denotation, independently of their morphological number (singular vs. plural). In Jackendoff's (1991) terms, singular country names behave conceptually as Groups [+bounded, +internal structure]: they are bounded entities (hence morphologically singular) but inherently possess internal partition. The syntactic reflex of this internal complexity ([+i]) is the obligation to Merge in N and project a D-layer to act as an argument. Striking evidence for the internal complexity of singular country names is their ability to be partitioned via modification (e.g., *il Portogallo settentrionale* / *il Portogallo del nord*, 'Northern Portugal'). This property is unavailable to true atomic, individual names (**la Roma meridionale* / **la Roma del sud*, 'Southern Rome'); because a city name is a punctual entity; identifying a sub-part of it requires the formation of a new bare, atomic proper name, such as *Roma Sud* ('South Rome').

⁶ Matushansky openly disagrees with both Fara and Jeshion regarding the syntactic derivation of such items, arguing that they cannot be simply treated as common count nouns derived from singular proper names.

⁷ It is crucial to emphasize that this morphosyntactic behavior is not an artifact of foreign loanword adaptation, but systematically applies to native Italian band names, overriding any internal anomaly of the proper root. For instance, the obligatory plural determiner and plural agreement pattern is perfectly replicated with Italian singular adjective-noun combinations (*Massimo Volume*), full declarative clauses (*Non voglio che Clara*), and acronyms (*CCCP*). Furthermore, the syntax strictly follows the biological gender of the aggregate's members even when it overrides highly deceptive native morphological features. A striking case is the Italian indie band *Marlene Kuntz*. Although the band name is morphologically composed of a female given name and a surname, the group comprises men only. Consequently, it obligatorily takes the masculine plural determiner, as shown in (i):

- (i) I Marlene Kuntz sono/*è un gruppo di Cuneo.
the.M.PL Marlene Kuntz are/is a band from Cuneo
'Marlene Kuntz are a band from Cuneo.'

⁸ It is also worth noting that this group-denoting origin provides a principled explanation for the obligatory definite determiner used with famous Italian surnames (e.g., *il Correggio*, *il Petrarca*, *il Boccaccio*). Family names inherently function as collective names referring to groups of persons. Because they denote a set rather than an atomic individual, they are structurally merged in N. When applied to designate a specific member of that family, the root maintains its set-denoting architecture and requires a D-layer to achieve referentiality. The definite determiner in these cases acts as a marker for the "dimension of knowledge," signalling to the hearer that the referent belongs to the domain of universal or encyclopedic knowledge rather than specific/local acquaintance.

⁹ Jeshion (2015), for instance, groups various modified names into categories such as "Producer Examples" (e.g., *a Picasso*) or "Representational Examples", treating them as instances of polysemy or semantic type-shifting rather than pure predication.

¹⁰ This mereological approach to adpositions builds on Rugna and Franco's (2022) analysis of Italian Prepositional Compounds. They demonstrate that prepositions linking two nominals are not semantically vacuous but act as elementary relators expressing different directionalities of part-whole relations. Specifically, they show that the preposition *da* lexicalizes an inclusion relator ($\subseteq$) (e.g., in *canna da pesca*, 'fishing rod', the rod is zonally included in the fishing event), while *a* lexicalizes a reverse inclusion relator ($\supseteq$) (e.g., in *barca a vela*, 'sailboat', the sail is included in the boat), functioning exactly like the instrumental/comitative preposition *con* ('with'). Crucially, they propose that *di* ('of') lexicalizes a fusion relator ($\otimes$). In compounds like *succo di limone* ('lemon juice') or idiomatic ones like *faccia di bronzo* ('bronze face'), *di* signals a single intersected zonal domain fusing the relevant properties of both the head and the modifier into a unified whole, without conceptual partitioning.

¹¹ It is crucial to clarify that our adoption of formal ‘mereotopological’ operators ($\otimes$, $\subseteq$, $\supseteq$) to model the syntax of materials and complex names does not imply that speakers possess explicit or conscious knowledge of formal ontology. Rather, in line with the Biolinguistic enterprise and research on the space-language cognitive interface, we assume that these operators formalize an innate, pre-theoretical cognitive instinct for part-whole and spatial relations (cf. Adger 2025). Natural language syntax co-opts the neurobiological capacity that the human brain evolved to parse the physical world — such as visual objecthood, boundaries, and containment — in order to structure nominal reference. In this sense, the grammar of materials and complex proper names is a direct reflex of how human cognition maps fundamental topological primitives onto the syntactic spine, a view entirely compatible with the core intuition of Longobardi’s (2005) *Topological Mapping Theory*. In doing so, our approach aligns with the framework of ‘commonsense metaphysics’ (Asher & Pustejovsky 2005) and the intersection of formal ontology and cognitive science (Smith 1995), where mereotopological primitives are taken to model the naïve, pre-theoretical conceptualization of the physical world inherent in human cognition.

¹² The concept of “fusion” or mereological product must be understood within the formal framework of mereotopology (Casati & Varzi 1999). While pure mereology studies the abstract part-whole relation, it is topologically blind: it cannot account for spatial continuity, boundaries, or physical contact. Topology introduces the primitive of “connection” (C) to solve this issue. By combining them, mereotopology can precisely model how spatial or material entities interact. In this context, a mereotopological fusion (or product) is not a simple set-theoretic intersection of abstract members; rather, it defines a topological operation where two connected domains overlap and merge their boundaries. The result is a single, unpartitioned, self-connected spatial or conceptual whole. This topological dimension formally captures how a material property (like *marble*) and a host entity (like a *statue*) are intrinsically fused together in the physical world, erasing any internal boundary between them.

¹³ This individual vs. aggregate distinction is extremely robust and extends systematically across all complex names. As documented in the literature, when the modifying proper root is an atomic individual, the intersection relator *di* obligatorily takes a bare NP, yielding widespread toponyms such as *Lago di Vico*, *Lago di Bolsena*, or *Mare di Alboran*. Conversely, when the modifier is conceptualized as a plural set or aggregate, it undergoes Merge in N and must project a full DP layer, forcing the appearance of the plural definite article. This is evident in famous non-geographical complex names like the ballet *Il Lago dei cigni* (‘Swan Lake’), but crucially also in genuine geographical toponyms such as *Il Lago delle ondine* (a small salt lake in Pantelleria) or *Il Golfo dei Poeti* (‘Gulf of Poets’ in Liguria). In all these cases, the plural aggregate nature of the modifier (*cigni*, *ondine*, *poeti*) structurally imposes the *dei/delle* (*di* plus the plural definite determiner *i/le*) articulation.

¹⁴ One might wonder why mass materials (e.g., *bronze* in *faccia di bronzo*) behave syntactically like atomic individuals (e.g., *Garda*) — merging as bare NPs under the $\otimes$ relator — rather than projecting a full DP like aggregate proper names. This asymmetry is fully predicted by Jackendoff’s (1991) nominal mereology. Both atomic individuals and continuous substances share the negative mereological feature [–internal structure] ([–i]), as they lack discrete, distinguishable internal partition. Because they lack internal members, they do not require a functional D-layer to bind or maximize a set. Conversely, group names like *My Bloody Valentine* are inherently [+internal structure] ([+i]); because they denote collections of distinguishable parts (the band members), they must project the D-layer to act as referential arguments.

¹⁵ While *-ico* can occasionally encode reverse inclusion with liquid mixtures, such as *bevanda alcolica* ‘alcoholic beverage / drink containing alcohol’, its application to solid materials systematically yields a whole-material fusion reading.

¹⁶ This structural derivation aligns with the insights of Anderson (2004, 2007), who argues that proper names do not belong to the lexical category of common nouns (N), but rather pattern with pronouns and determiners as ‘determinatives’. Following this rationale (cf. also Burnett 2006), it would be paradoxical to assume that an inherently individual-denoting root must first undergo Merge in N only to subsequently raise to D. Because atomic proper roots are intrinsically individuated, they are base-generated directly in the D-layer, lacking the nominal structure that would require further functional determination.

¹⁷ Franco and Rugna (2024) show that while unmodified individual proper names are hosted directly in D (or undergo N-to-D movement) to secure rigid designation, the introduction of attributive modifiers fundamentally alters this architecture. Franco and Rugna (2024) show that modified proper names, such as *una piccola Firenze* (‘a little Florence’) or *la Firenze del Baltico* (‘the Florence of the Baltic’), are obligatorily merged in N. Because they undergo Merge in N, they are stripped of their inherent referential autonomy and behave morphosyntactically like standard common count nouns. Consequently, the syntax must project a full DP layer and recruit an overt determiner (either definite or indefinite) to close the nominal expression and enable it to function as a referential argument. This provides evidence that the emergence of the definite article in expressions like *la bella Firenze* (‘the beautiful Florence’) is not an

idiosyncratic exception, but the regular morphosyntactic reflex of a proper root being ‘trapped’ in N by the presence of a modifier.

¹⁸ The semantic restriction barring aggregates from merging directly in D stems from their internal mereological complexity. As we have already pointed out, unlike atomic individuals, which possess intrinsic referential autonomy and pattern universally with pronouns as ‘determinatives’ (Anderson 2004), aggregate proper names denote aggregates or collections inherently characterized by ‘weak individuality’ (Acquaviva 2008). Following Link (1983) and Chierchia (1998), plurals and mass-like aggregates require functional mediation to establish a fixed reference. Therefore, they must be base-generated in N so that a higher functional operator (the D-layer) can maximize the set and convert it into a fully referential argument. As highlighted by Borer (2005), the structural partitioning of an entity operates fundamentally on the nominal projection. Consequently, forcing an aggregate to undergo Merge directly in D would treat an internally complex mereotopological expanse as a dimensionless atom, thereby violating the basic mapping constraints between ontology and syntax.

¹⁹ We must carefully distinguish the dual mereosyntactic nature of the relator *di* (‘of’). In standard possessive or part-whole constructions (e.g., *il cane dei ragazzi*, ‘the boys’ dog; *le proboscidi degli elefanti*, ‘the elephants’ trunks’), *di* spells out an inclusion relator ($\subseteq$), indicating that the possessum is structurally included within the possessor’s broader zonal domain. Conversely, as already pointed out, in whole-material and toponymic constructions (e.g., *faccia di bronzo*, ‘bronze face’; *Lago di Garda*, ‘Lake Garda’; *succo di limone*, ‘lemon juice’), the relation is entirely non-possessive and the mereological product $\otimes$ isolates the maximal common overlapping part of two spatial, conceptual, or material domains.

²⁰ Cross-linguistic evidence further refutes the Predicativist assumption that proper names are merely defective common nouns. Looking at the inflectional paradigms of Modern Greek, Marinis (2025) demonstrates that toponyms exhibit ‘anti-defective’ phenomena, circumventing the morphological restrictions of standard nouns. Crucially, Marinis shows that toponyms structurally diverge not only from common nouns but also from anthroponyms (being strictly bound to singularia or pluralia tantum paradigms), providing robust morphological backing to our claim that spatial aggregates and atomic individuals possess fundamentally different morphosyntactic architectures.

²¹ Evidence that proper names and common nouns project different syntactic structures—and therefore cannot be uniformly analyzed as predicates—comes from alignment systems and agreement overrides. In Kalaw Lagaw Ya (an Australian language of the Pama-Nyungan family), common nouns and proper names exhibit completely opposite case alignment systems. While common nouns follow an Ergative-Absolutive pattern (ia), proper names operate on a Nominative-Accusative system (ib) (Comrie 1981: 8).

- | | | | | |
|-----|----|------------------------|--------------------|-----------|
| (i) | a. | Garkoez-in | burum- $\emptyset$ | mathaman. |
| | | man-ERG | pig-ABS | hit |
| | | ‘The man hit the pig.’ | | |
| | b. | Kala- $\emptyset$ | Gibuma-n | mathaman. |
| | | Kala-NOM | Gibuma-ACC | hit |
| | | ‘Kala hit Gibuma.’ | | |
- Kalaw Lagaw Ya*

If proper names were structurally identical to common count nouns, such a radical split in case assignment would be unexpected. Further evidence that “properhood” is a distinct structural configuration (Merge in D) is found in Orungu, a Bantu language of Gabon (Van de Velde & Ambouroué 2011). In Orungu, proper names are frequently derived directly from the common noun lexicon. For instance, the lexical root *òkéró*, which literally translates to ‘a part’, natively functions as a common noun belonging to noun class 3. However, when this root is merged into the syntax as an anthroponym (specifically, a woman’s name), it sheds its original morphological behavior. Instead of triggering Class 3 agreement, the proper name *Okeró* adopts a Definite Tone Pattern (DTP) — reflecting its inherent referentiality without the need for a structural determiner — and triggers class 9 agreement on modifiers such as demonstratives. This morphological shift shows that a proper name does not inherit the syntactic properties of the common noun it derives from. Rather, the direct merge of the root into a referential position overrides its original nominal syntax, imposing a specific ‘onymic’ agreement. This shows that referential proper names are not just count nouns embedded under a null determiner: their direct merge into a referential position aligns them with the highly individuated status of personal pronouns on the Animacy and Definiteness Hierarchies (Salaberri 2022; Ortmann 2014).

²² In Ancient Greek and Latin, toponyms frequently utilize bare case-forms (e.g., bare Locative/Dative for location and bare Accusative for goal) in contexts where common nouns would obligatorily require prepositional phrases (Stolz, Lestrade & Stolz 2014; Haspelmath 2019). Similarly, in Basque, while common nouns obligatorily take class markers like -ta (inanimate) and -gan (animate) within their semantic case paradigms, toponyms completely lack these markers, making them morphologically far less complex (Trask 2003; De Rijk 2008). This cross-linguistic bareness is the direct morphological reflex of the proper root undergoing Merge in D (or raising to D): lacking internal mereological structure, the atomic name requires no further functional adpositional layers to be anchored in space or discourse. Furthermore, research into American Sign Language (ASL) demonstrates that “name signs” bypass standard syntactic

mechanisms applied to common nouns. While ASL heavily utilizes classifier predicates to denote classes of common nouns (structuring them spatially and dimensionally), arbitrary name signs (Supalla, 1990) resist integration into these complex classifier templates. Instead, they function as rigid, atomic lexical items that interact entirely differently with the spatial syntax of the language, functioning as pure rigid referents. This corroborates the claim that "properhood" is a distinct, deeply embedded cognitive structural configuration.

²³ This precise cross-linguistic asymmetry between Romance and Germanic architectures is systematically replicated in the morphosyntactic encoding of whole-material complements, providing striking independent evidence for our unified mereosyntactic approach. The mereological fusion of a sortal and a material substance obligatorily requires the overt prepositional relator *di* ($\otimes$) in Italian and other Romance languages (e.g., *casa di mattoni*, 'house of bricks'; *statua di bronzo*, 'statue of bronze'). Conversely, Germanic languages regularly encode this exact same whole-material relation through nominal compounding or Left-Branch N-N juxtaposition, entirely lacking adpositional relators (e.g., English *brick house* or *bronze statue*; German *Steinhaus*, 'stone house'). The fact that complex toponyms and whole-materials exhibit the exact same parametric variation strongly corroborates the hypothesis that the syntactic linkage of a proper root to a sortal is not an idiosyncratic onomastic rule, but is governed by the universal mereosyntactic primitives regulating material constitution.

²⁴ The necessity of treating river names as unbounded, set-like entities rather than atomic individuals is extensively supported by the typological study of D'hulst, Thieroff, and Meisenburg (2022). They demonstrate that in all West-Germanic and Romance languages, river names obligatorily take the definite article. The authors argue that this behavior stems from a fundamental semantic property: unlike cities (which are conceptualized as punctual atoms) or even lakes (which are bounded by their shores), rivers are 'lexically underspecified for boundedness'. They denote a flowing, massively extended substance. Because they are unbounded and mereologically complex, they cannot achieve referential autonomy on their own. The syntactic underspecification must be resolved by the obligatory presence of the definiteness feature provided by the D-layer. This perfectly aligns with our theoretical architecture: possessing wide spatio-temporal extensions, rivers—just like states or mountain ranges—cannot undergo Merge in D. They merge in N as aggregates/expanses and obligatorily project an overt definite determiner.